

Empirical Literature Review Note: Housing Supply and Fertility (Economics-First, Causal Evidence Focus)

Project 03: Fertility and Housing Supply

February 20, 2026

Scope and rule

This note focuses on empirical economics evidence with attention to causal identification. It is based on verified local PDFs in:

- `projects/03_Fertility_and_Housing_Supply/Literature/`

Theoretical models are intentionally deferred to a follow-up note.

1 Bottom line for your concern (weak empirical footing)

The strongest direct fertility-housing empirical evidence in the current local folder is **Couillard (2025, JMP)**. Most other high-quality empirical papers in the folder identify the housing *supply-to-price* link rather than *price-to-fertility* directly.

So the empirical chain is currently:

1. supply constraints/upzoning $\rightarrow$ prices/quantities (strong evidence),
2. prices/composition $\rightarrow$ fertility (promising but thinner in this local set).

2 Empirical economics evidence (causal design first)

A. Direct housing–fertility evidence

Couillard (2025), “Build, Baby, Build: How Housing Shapes Fertility” (JMP).

From the abstract in the local draft: the paper estimates a dynamic joint housing-fertility model using U.S. Census Bureau data, allows location and unit-size choices, and runs policy/supply counterfactuals. Reported magnitudes include: housing-cost increases since 1990 explaining 11% fewer children, 51% of the 2000s-to-2010s TFR decline, and 7 percentage points fewer young families; supply shifts for large units produce 2.3x the births of equal-cost shifts for small units.

Identification assessment: this is structural causal quantification (model-based counterfactual variation in costs/supply), not a reduced-form natural experiment. It is directly relevant to your project because unit size enters as a central mechanism.

B. Housing supply constraints and price transmission (upstream causal block)

Saiz (2010, QJE).

Uses predetermined geographic constraints (terrain and water) to characterize local supply elasticities and shows strong heterogeneity in supply responsiveness across U.S. metros.

Identification assessment: quasi-exogenous geography provides credible variation for supply constraints.

Hilber and Vermeulen (2012 working paper version in local folder).

Panel of English local planning authorities (1974–2008). Uses policy reform variation, vote shares, and historical density as instruments for endogenous local constraints. Finds constraints amplify house-price responses to demand and reports large long-run price effects.

Identification assessment: explicit IV strategy targeting endogeneity of regulatory constraints.

Freemark (2019, Urban Affairs Review).

Difference-in-differences around Chicago upzoning that increased allowed density and reduced parking requirements in ways described as exogenous to neighborhood development plans. Finds higher land/condo prices but no significant increase in permitted units over five years.

Identification assessment: local policy quasi-experiment with transparent DiD design; informative about implementation frictions and short-run quantity effects.

C. Demographic pressure and prices (reverse-direction benchmark)

Francke and Korevaar (2022, SSRN draft in local folder).

Documents that birth-rate variation predicts house prices decades later (1 pp higher birth rate associated with 4–5% higher prices 25–30 years later), with limited effects on rents.

Identification assessment: strong long-horizon reduced-form evidence on demography → house prices. For your project, this is useful as a dynamic background fact, not direct evidence of prices → fertility.

3 What this implies for the NIMBY-fertility paper

1. You can already claim strong empirical support for the **supply-constraint-to-price** link from Saiz, Hilber–Vermeulen, and Freemark.
2. You now have a directly relevant fertility paper (Couillard) linking **housing affordability/composition to fertility outcomes** with structural causal counterfactuals.
3. The remaining empirical vulnerability is a shortage of reduced-form causal papers that cleanly estimate **housing-cost shocks on fertility timing/completed fertility**. [INFERRED – based on current local folder contents]

4 Brief note on non-economics literatures

There is broader work in sociology/demography/policy on housing affordability and delayed family formation, but this note does not rely on it because your criterion is rigorous causal economics evidence.

5 Verification table

Paper	Key claim (from source)	Relevance to current paper	Confidence
Couillard (2025)	Housing costs and housing-unit composition materially affect fertility in structural counterfactuals; large-unit supply shifts produce larger fertility responses.	Direct evidence for the core mechanism (housing costs/supply composition $\rightarrow$ fertility).	High
Saiz (2010)	Geography strongly shapes local housing supply elasticity.	Provides causal/credible upstream supply elasticity block for your mechanism.	High
Hilber & Vermeulen (2012 WP)	Regulatory constraints causally amplify house-price responses to demand in England.	Directly useful for UK-style regulatory channel from planning constraints to affordability.	High
Freemark (2019)	Upzoning raised values but did not raise permitted units in short run.	Shows that easing regulation may have delayed quantity effects; important for transitional dynamics in empirical motivation.	High
Francke & Korevaar (2022)	Birth-rate changes predict future house-price cycles.	Useful dynamic complement; highlights two-way demography-housing interactions.	Medium

6 Next empirical priority (before theory note)

To tighten the empirical section further, add 2–4 reduced-form papers that estimate fertility responses to plausibly exogenous housing-cost shocks, then separate your evidence discussion into:

- **Direct fertility effects** (price/rent/supply shocks on births),
- **Upstream housing-supply effects** (regulation/geography/policy on prices and unit mix).